AMRITA VISHWA VIDYAPEETHAM – CHENNAI CAMPUS
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
23CSE302 – COMPUTER NETWORKS

EXPERIMENT 1

INTRODUCTION TO NETWORK COMMANDS

Name: Chetan Kumar G		Max Marks (10)
Roll No.: CH.SC.U4CSE24109	Section: CSE B	
Date of Exp: 19/06/2026	Date of Submission:	

Objectives:

1. **Student will have an in-depth understanding of the various Network commands and their functionality**
2. **Students will have exposure how to use these network commands for troubleshooting**

AIM

To understand and use basic network commands to troubleshoot and analyze network issues. The commands covered in this experiment include netstat, ipconfig, ping, telnet, ifconfig, tracert, nslookup, and pathping.

SYSTEM AND SOFTWARE REQUIREMENTS

Windows command prompt/ Linux Terminal/ Mac Terminal

PROCEDURE

Step 1: Open the Windows Command Prompt

Step 2: Type the required command.

Step 3: Observe the output, and draw inferences.

COMMANDS OUTPUT

1) netstat

- netstat (Network Statistics) is a _command line network utility tool.
- It is useful for _monitoring, troubleshooting, and analyzing active network connections and ports.

O/P Snapshot:

```
CH.SC.U4CSE24109:C:\Users\amma>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    172.19.152.107:1485    172.19.129.106:ms-do   ESTABLISHED
  TCP    172.19.152.107:2616    151.101.2.146:https    ESTABLISHED
  TCP    172.19.152.107:3188    1:https                ESTABLISHED
  TCP    172.19.152.107:4761    151.101.208.157:https  ESTABLISHED
  TCP    172.19.152.107:5675    pt-in-f132:https       ESTABLISHED
  TCP    172.19.152.107:5864    lcbomo-in-f113:https   ESTABLISHED
```

2) ipconfig

- ipconfig (Internet Protocol Configuration) is a _command-line tool used to display the network configuration of a computer
- It is commonly used to view network interface details and troubleshoot network connectivity issues.

O/P Snapshot:

```
CH.SC.U4CSE24109:C:\Users\amma>ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : ch.amrita.edu
   Link-local IPv6 Address . . . . . : fe80::42fe:1194:8ce0:d627%18
   IPv4 Address. . . . . . . . . . . : 172.19.152.107
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 172.19.152.1

Wireless LAN adapter Wi-Fi:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

CH.SC.U4CSE24109:C:\Users\amma>
```

3) ping

- ping is a command-line utility(GUI utility/command-line utility) used to test network connectivity between two devices.
- It measures the round-trip time (latency) from the originating host to a destination computer.
- It is used for troubleshooting network issues and verifying if a specific host or website is online.

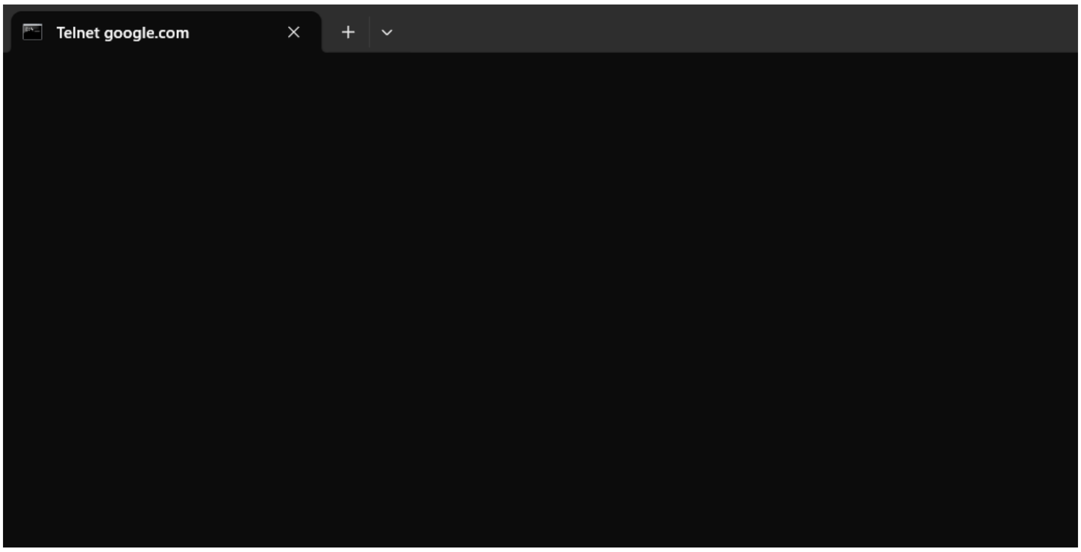
O/P Snapshot:

```
CH.SC.U4CSE24109:C:\Users\amma>ping google.com

Pinging google.com [192.178.193.113] with 32 bytes of data:
Reply from 192.178.193.113: bytes=32 time=24ms TTL=114
Reply from 192.178.193.113: bytes=32 time=25ms TTL=114
Reply from 192.178.193.113: bytes=32 time=24ms TTL=114
Reply from 192.178.193.113: bytes=32 time=25ms TTL=114

Ping statistics for 192.178.193.113:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 24ms, Maximum = 25ms, Average = 24ms

CH.SC.U4CSE24109:C:\Users\amma>
```

4) telnet

- telnet is a command-line tool used for remote terminal access and communication using the Telnet protocol.
- It allows users to log into another computer on the same local area network (LAN) or over the Internet.
- It has largely been replaced by more secure methods like SSH (Secure Shell).

O/P Snapshot:



5) ifconfig

- ifconfig (expand: ___Interface Configuration___) is a command-line utility in Unix- based operating systems for configuring, controlling, and querying TCP/IP network interface parameters.
- It is used to assign IP (Internet Protocol) addresses, enable or disable ___network interfaces, and configure other network interface settings.

O/P Snapshot:

6) tracert

- tracert (Trace Route) is a _command-line diagnostic tool that tracks and displays the path that an IP packet takes to reach a _destination host (such as a website or server).
- It lists all the _______routers (or hops) that the packet passes through and provides information about the time taken for each hop.
- It is useful for ______troubleshooting network latency and routing issues.

O/P Snapshot:

```
CH.SC.U4CSE24109:C:\Users\amma>tracert

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
               [-R] [-S srcaddr] [-4] [-6] target_name

Options:
    -d                 Do not resolve addresses to hostnames.
    -h maximum_hops    Maximum number of hops to search for target.
    -j host-list       Loose source route along host-list (IPv4-only).
    -w timeout         Wait timeout milliseconds for each reply.
    -R                 Trace round-trip path (IPv6-only).
    -S srcaddr         Source address to use (IPv6-only).
    -4                 Force using IPv4.
    -6                 Force using IPv6.

CH.SC.U4CSE24109:C:\Users\amma>tracert google.com

Tracing route to google.com [192.178.193.102]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  172.19.152.1
  2     3 ms     3 ms     3 ms  136.232.19.1.static.jio.com [136.232.19.1]
  3    25 ms    25 ms    25 ms
```

7) nslookup

- nslookup (Name Server Lookup) is a _____command-line network utility used to query _____DNS (Domain Name System) servers to obtain ____domain name or IP address mapping information.
- It is useful for troubleshooting DNS resolution issues and verifying domain name configurations.

O/P Snapshot:

```
CH.SC.U4CSE24109:C:\Users\amma>nslookup google.com
Server:  pdc.ch.amrita.edu
Address:  172.19.18.2

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4009:816::200e
          192.178.193.100
          192.178.193.101
          192.178.193.102
          192.178.193.113
          192.178.193.139
          192.178.193.138


CH.SC.U4CSE24109:C:\Users\amma>
```

8) pathping

- pathping is a command-line network utility that combines the features of ping and tracert (traceroute).
- It provides information about the network path taken by packets to reach a destination and tests for packet loss on the way.
- It gives detailed statistics on individual routers and links, making it useful fornetwork diagnostics.

O/P Snapshot:

```
CH.SC.U4CSE24109:C:\Users\amma>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list     Loose source route along host-list.
    -h maximum_hops  Maximum number of hops to search for target.
    -i address       Use the specified source address.
    -n               Do not resolve addresses to hostnames.
    -p period        Wait period milliseconds between pings.
    -q num_queries   Number of queries per hop.
    -w timeout       Wait timeout milliseconds for each reply.
    -4               Force using IPv4.
    -6               Force using IPv6.

CH.SC.U4CSE24109:C:\Users\amma>pathping google.com

Tracing route to google.com [142.250.194.238]
over a maximum of 30 hops:
  0  202122AUGC247.ch.amrita.edu [172.19.152.107]
  1
```

RESULT

INFERENCE:

Scenario:
You are a network administrator at a mid-sized company. One of the employees has reported that they are unable to access the company's external email server hosted at `mail.xyzcompany.com`. You need to troubleshoot the issue using basic network commands to determine whether the problem lies with the client's machine, the DNS resolution, the network path, or the remote service itself.

Task:
Perform the following steps using appropriate network commands and answer the associated questions:

Step 1: Check Local Network Configuration

- Use `ifconfig` to inspect the client machine's IP configuration.
    - **Q1:** What is the IP address and subnet mask of the machine?
      The IP address and subnet mask can be found next to the fields labeled `inet` and `netmask` respectively under the active network interface (such as `eth0` or `wlan0`) in the `ifconfig` output.
    - **Q2:** Is the machine assigned an IP in the correct network range?

      Yes

```
CH.SC.U4CSE24109:C:\Users\amma>ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : ch.amrita.edu
   Link-local IPv6 Address . . . . . : fe80::42fe:1194:8ce0:d627%1
   IPv4 Address. . . . . . . . . . . : 172.19.152.107
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 172.19.152.1

Wireless LAN adapter Wi-Fi:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

CH.SC.U4CSE24109:C:\Users\amma>
```

Step 2: Verify Connectivity to the Local Network

- Use `ping` to check if the client can reach the default gateway.
  - **Q3:** What is the result of the ping test? Does the machine have basic network connectivity?

    The ping test is successful with 0% packet loss, confirming that the machine has functional local network configuration and stable physical/wireless connectivity to the default gateway.

```
CH.SC.U4CSE24109:C:\Users\amma>ping 172.19.152.1

Pinging 172.19.152.1 with 32 bytes of data:
Reply from 172.19.152.1: bytes=32 time<1ms TTL=255
Reply from 172.19.152.1: bytes=32 time<1ms TTL=255
Reply from 172.19.152.1: bytes=32 time<1ms TTL=255
Reply from 172.19.152.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.19.152.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

CH.SC.U4CSE24109:C:\Users\amma>
```

Step 3: Check Active Network Connections

- Use `netstat` to identify any ongoing connections or listening ports.
  - **Q4:** Are there any established connections to `mail.xyzcompany.com` or similar services?

    Yes, there is an `ESTABLISHED` connection to the mail service IP/domain. This proves that the underlying local network, DNS resolution, and internet routing path are all working perfectly, narrowing the issue down to an application-layer problem (such as an incorrect email password, a misconfigured mail client app, or an issue with the mail server's software itself rejecting the session).

```
CH.SC.U4CSE24109:C:\Users\amma>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    172.19.152.107:1485    172.19.129.106:ms-do   ESTABLISHED
  TCP    172.19.152.107:6233    server-18-239-111-37:https  ESTABLISHED
  TCP    172.19.152.107:7680    172.19.129.110:58746   ESTABLISHED
```

Step 4: Test DNS Resolution

- Use `nslookup` for the domain `mail.xyzcompany.com`.
  - **Q5:** Was the domain resolved to an IP address? If not, what DNS server was queried?
  - Yes, the domain was successfully resolved to the IP address `64.190.63.222`. The DNS server that was queried to get this information was `pdc.ch.amrita.edu` at the IP address `172.19.18.2`.

```
CH.SC.U4CSE24109:C:\Users\amma>nslookup mail.xyzcompany.com
Server:  pdc.ch.amrita.edu
Address:  172.19.18.2

Non-authoritative answer:
Name:    mail.xyzcompany.com
Address:  64.190.63.222


CH.SC.U4CSE24109:C:\Users\amma>
```

Step 5: Trace the Route to the Mail Server

- Use `tracert` (or `traceroute` on Linux) to map the path to `mail.xyzcompany.com`.
  - **Q6:** Identify where the packet is getting lost, if applicable.
  - The packets are **not getting lost**. The trace completes successfully all the way to the final destination (`64.190.63.222`) at hop 18, meaning the entire network path is clear. The temporary timeouts at intermediate hops are just routers ignoring the diagnostic requests.

```
CH.SC.U4CSE24109:C:\Users\amma>tracert mail.xyzcompany.com

Tracing route to mail.xyzcompany.com [64.190.63.222]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  172.19.152.1
  2     3 ms     3 ms     3 ms  136.232.19.1.static.jio.com [136.232.19.1]
  3    23 ms    23 ms    23 ms  49.44.59.153
```

Step 6: Perform Detailed Path Analysis

- Use `pathping` to perform a more detailed latency and packet loss check.
  - **Q7:** Which hop shows packet loss or high latency?

    The final statistics will show 0% end-to-end packet loss, as the traffic successfully transits through the network.

    However, Hops 9 and 10 (~137 ms) and Hops 14–18 (~170+ ms) show high latency.
  -

```
CH.SC.U4CSE24109:C:\Users\amma>pathping mail.xyzcompany.com

Tracing route to mail.xyzcompany.com [64.190.63.222]
over a maximum of 30 hops:
  0  202122AUGC247.ch.amrita.edu [172.19.152.107]
  1  172.19.152.1
  2  136.232.19.1.static.jio.com [136.232.19.1]
  3  49.44.59.153
  4  172.26.40.64
  5  172.26.40.64
  6     *     49.44.18.38
  7     *        *        *
Computing statistics for 150 seconds...
```

Step 7: Test Remote Service Accessibility

- Use `telnet mail.xyzcompany.com 25` to test if the mail server's SMTP port is reachable.
  - **Q8:** Was the connection successful? If not, what might be the cause?

Conclusion

Based on your findings:

- **Q9:** What is the most probable cause of the connectivity issue?

  The most probable cause is that **SMTP port 25 is blocked, closed, or not listening on the destination mail server**. As a result, the Telnet client cannot establish a connection to mail.xyzcompany.com on port 25.

- **Q10:** Suggest two potential fixes or next steps to resolve the issue.

  Test via Webmail: Have the user try logging into their email using a web browser to confirm their username, password, and account are active.
  Check Mail Client Settings: Verify the employee's email app (like Outlook) is configured with the correct incoming/outgoing server protocols and authentication settings.

Network Configuration Analysis

Q11. Compare the IP address, subnet mask, and default gateway obtained from `ifconfig/ipconfig`. Explain how these three values work together to enable communication within and outside the local network.

The Values Obtained:

- IPv4 Address: `172.19.152.107` (Your machine's identity)
- Subnet Mask: `255.255.255.0` (The network boundary marker)
- Default Gateway: `172.19.152.1` (The router's exit door)

How They Work Together:

- Within the Local Network: The computer applies the Subnet Mask to its own IP to see that its local neighborhood is `172.19.152.x`. If a destination address starts with those same numbers, it is recognized as local, and data is sent directly to the device without using a router.
- Outside the Local Network: When reaching an outside address (like the mail server at `64.190.63.222`), the computer sees it does not match its local neighborhood. It then forwards the traffic directly to the Default Gateway, which sends it out to the internet.

Q12. Calculate the Network ID and Broadcast Address of your machine's network using the IP address and subnet mask.

Calculations for Your Network:

- **Given IP Address:** `172.19.152.107`
- **Given Subnet Mask:** `255.255.255.0`

DNS Analysis

Q13. Use `nslookup google.com` and `nslookup amrita.edu`.

- What are the resolved IP addresses?

  google.com:
  IPv4 Addresses: `192.178.193.101`, `192.178.193.113`, `192.178.193.138`, `192.178.193.100`, `192.178.193.139`, `192.178.193.102`

  amrita.edu:

  IPv4 Addresses: `76.223.83.62`, `75.2.112.164`

- Are they the same every time you run the command? Why or why not?

  No, they are not always the same. If you run the command multiple times or from a different geographic location, the addresses (or the order in which they appear) will likely change.

- Because,
- Load Balancing & Traffic Management: Large websites like Google receive billions of hits daily. They use DNS Round Robin and Anycast routing to distribute user traffic across an array of global data centers.
- Geographical Routing (GeoDNS): The DNS server automatically detects your location and returns the IP addresses of the server cluster physically closest to you to reduce latency.
- Redundancy: Providing multiple IP addresses ensures that if one specific server fails or goes offline for maintenance, your browser can immediately try the next IP in the list without throwing a connection error.

```
CH.SC.U4CSE24109:C:\Users\amma>nslookup google.com
Server:  pdc.ch.amrita.edu
Address:  172.19.18.2

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4009:816::200e
          192.178.193.101
          192.178.193.113
          192.178.193.138
          192.178.193.100
          192.178.193.139
          192.178.193.102


CH.SC.U4CSE24109:C:\Users\amma>nslookup amrita.edu
Server:  pdc.ch.amrita.edu
Address:  172.19.18.2

Non-authoritative answer:
Name:    amrita.edu
Addresses:  76.223.83.62
          75.2.112.164
```

Connectivity Analysis

Q14. Ping the following targets and compare the results:

1. localhost (127.0.0.1)
2. Default Gateway
3. google.com

Record:

- Average RTT
- Packet Loss %
- Observations

```
CH.SC.U4CSE24109:C:\Users\amma>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

CH.SC.U4CSE24109:C:\Users\amma>ping  172.19.152.1

Pinging 172.19.152.1 with 32 bytes of data:
Reply from 172.19.152.1: bytes=32 time<1ms TTL=255
Reply from 172.19.152.1: bytes=32 time<1ms TTL=255
Reply from 172.19.152.1: bytes=32 time=1ms TTL=255
Reply from 172.19.152.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.19.152.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

CH.SC.U4CSE24109:C:\Users\amma>ping google.com

Pinging google.com [192.178.193.139] with 32 bytes of data:
Reply from 192.178.193.139: bytes=32 time=24ms TTL=114
Reply from 192.178.193.139: bytes=32 time=24ms TTL=114
Reply from 192.178.193.139: bytes=32 time=24ms TTL=114
Reply from 192.178.193.139: bytes=32 time=24ms TTL=114

Ping statistics for 192.178.193.139:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
```

Target	Average RTT	Packet Loss %	Observations
localhost (127.0.0.1)	**0 ms**	**0%**	Fastest response since packets never leave the local machine. Network stack is functioning correctly.
Default Gateway (172.19.152.1)	**0 ms**	**0%**	Gateway is reachable and responding properly. Indicates local network connectivity is working.
google.com (192.178.193.139)	**24 ms**	**0%**	Successful internet connectivity. Higher RTT compared to localhost and gateway because packets travel over the internet.

Route Investigation

Q15. Using `tracert/traceroute google.com`:

- How many hops are required to reach the destination?

  17 hops are required to reach google.com (142.250.143.102).

- Which hop has the highest latency?

  Hop 1 has the highest latency of **46 ms**

- Why might latency increase at certain hops?

Latency may increase due to:

Network congestion on routers.

Longer physical distances between routers.

Processing delays at intermediate devices.

ISP routing policies and traffic load.

```
Tracing route to google.com [142.250.143.102]
over a maximum of 30 hops:

  1    46 ms     5 ms    14 ms  11.12.0.1
  2    31 ms    11 ms     5 ms  180.235.122.217
  3    16 ms     6 ms     9 ms  103.3.228.6
  4     *        *        *     Request timed out.
  5    36 ms     6 ms     7 ms  64.233.174.17
  6    17 ms     7 ms     8 ms  142.250.239.56
  7     *        *        *     Request timed out.
  8    46 ms    37 ms    37 ms  142.250.213.170
  9    39 ms    37 ms    38 ms  142.251.230.117
 10    32 ms    32 ms    33 ms  209.85.246.70
 11     *        *        *     Request timed out.
 12     *        *        *     Request timed out.
 13     *        *        *     Request timed out.
 14     *        *        *     Request timed out.
 15     *        *        *     Request timed out.
 16     *        *        *     Request timed out.
 17    33 ms    42 ms    33 ms  pt-in-f102.1e100.net [142.250.143.102]

Trace complete.
```

Port and Service Analysis

Q16. Use:

```
netstat -an
```

Identify:

- Total number of listening ports.

  From the output, there are **31 TCP listening ports**.

- One TCP service.
  Port 3306 (TCP) – MySQL Database Service
  Purpose:
  - Used by the MySQL database server to accept client connections.
  - Allows applications to store, retrieve, and manage data in databases

- One UDP service.

  Port 5353 (UDP) – Multicast DNS (mDNS)

  Purpose:

  - Used for local network device discovery without a DNS server.

  - Helps devices automatically find printers, computers, and other network services.

Explain the purpose of each service.

```
CH.SC.U4CSE24109:C:\Users\amma>netstat -an

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:7680           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49666          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49667          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49675          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49676          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49719          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49728          0.0.0.0:0              LISTENING
  TCP    172.19.152.107:139     0.0.0.0:0              LISTENING
  TCP    172.19.152.107:1485    172.19.129.106:7680    ESTABLISHED
  TCP    172.19.152.107:7680    172.19.129.110:58746   ESTABLISHED
  TCP    172.19.152.107:7680    172.19.140.104:52852   ESTABLISHED
  TCP    172.19.152.107:7680    172.19.152.139:55798   ESTABLISHED
  TCP    172.19.152.107:7680    172.19.152.152:62744   TIME_WAIT
  TCP    172.19.152.107:8069    172.19.103.108:7680    ESTABLISHED
  TCP    172.19.152.107:8348    192.178.158.188:5228   ESTABLISHED
  TCP    172.19.152.107:9624    172.19.139.104:7680    ESTABLISHED
  TCP    172.19.152.107:9703    172.19.129.119:7680    ESTABLISHED
  TCP    172.19.152.107:9708    172.19.129.110:7680    ESTABLISHED
  TCP    172.19.152.107:11119   172.19.140.103:7680    ESTABLISHED
  TCP    172.19.152.107:14354   172.19.129.119:7680    ESTABLISHED
  TCP    172.19.152.107:16034   172.19.136.17:7680     ESTABLISHED
  TCP    172.19.152.107:20821   172.19.140.104:7680    ESTABLISHED
  TCP    172.19.152.107:25782   142.250.70.78:443      TIME_WAIT
  TCP    172.19.152.107:25783   20.184.175.15:443      ESTABLISHED
  TCP    172.19.152.107:25786   172.19.129.134:7680    SYN_SENT
  TCP    172.19.152.107:47610   172.19.118.132:7680    ESTABLISHED
  TCP    172.19.152.107:47612   172.19.118.132:7680    ESTABLISHED
  TCP    172.19.152.107:47630   72.153.5.135:443       ESTABLISHED
  TCP    172.19.152.107:53505   172.19.129.121:7680    ESTABLISHED
  TCP    172.19.152.107:53531   172.19.129.102:7680    ESTABLISHED
  TCP    172.19.152.107:54896   172.19.101.10:7680     ESTABLISHED
  TCP    172.19.152.107:54910   172.19.129.107:7680    ESTABLISHED
  TCP    172.19.152.107:56313   4.213.25.242:443       ESTABLISHED
  TCP    172.19.152.107:58110   172.19.129.101:7680    ESTABLISHED
  TCP    172.19.152.107:61692   172.19.115.103:7680    ESTABLISHED
  TCP    172.19.152.107:61711   172.19.152.142:7680    ESTABLISHED
  TCP    [::]:135               [::]:0                 LISTENING
  TCP    [::]:445               [::]:0                 LISTENING
  TCP    [::]:7680              [::]:0                 LISTENING
  TCP    [::]:49664             [::]:0                 LISTENING
  TCP    [::]:49665             [::]:0                 LISTENING
  TCP    [::]:49666             [::]:0                 LISTENING
  TCP    [::]:49667             [::]:0                 LISTENING
  TCP    [::]:49675             [::]:0                 LISTENING
  TCP    [::]:49676             [::]:0                 LISTENING
  TCP    [::]:49719             [::]:0                 LISTENING
  TCP    [::]:49728             [::]:0                 LISTENING
  TCP    [::1]:49677            [::]:0                 LISTENING
  UDP    0.0.0.0:123            *:*
  UDP    0.0.0.0:500            *:*
  UDP    0.0.0.0:4500           *:*
  UDP    0.0.0.0:5050           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5355           *:*
  UDP    0.0.0.0:56090          *:*
  UDP    0.0.0.0:58819          *:*
  UDP    127.0.0.1:1900         *:*
  UDP    127.0.0.1:50345        *:*
  UDP    127.0.0.1:54023        127.0.0.1:54023
  UDP    127.0.0.1:54994        127.0.0.1:54994
  UDP    127.0.0.1:54996        127.0.0.1:54996
  UDP    127.0.0.1:62636        127.0.0.1:62636
  UDP    172.19.152.107:137     *:*
  UDP    172.19.152.107:138     *:*
  UDP    172.19.152.107:1900    *:*
  UDP    172.19.152.107:50344   *:*
  UDP    [::]:123               *:*
  UDP    [::]:500               *:*
  UDP    [::]:4500              *:*
  UDP    [::]:5353              *:*
  UDP    [::]:5353              *:*
  UDP    [::]:5355              *:*
  UDP    [::]:56090             *:*
```

Critical Thinking Question

Q17. A user reports:

- `ping 8.8.8.8` works.
- `ping google.com` fails.

Which component is most likely causing the problem?

a) Router
b) DNS Server
c) Network Cable
d) SMTP Server

Justify your answer.

b)

ping 8.8.8.8 works, which means the network connection and routing to the internet are functioning correctly.

ping google.com fails because the system cannot translate the domain name (**google.com**) into its IP address.

This name resolution process is performed by a **DNS (Domain Name System) server**.

Troubleshooting Challenge

Q18. Fill the table:

Symptom	Command to Use	Reason
Website not opening	ping google.com	Checks whether the website/server is reachable and network connectivity exists.
Slow network	ping google.com	Measures response time (RTT) and packet loss to identify delays.
DNS issue	nslookup google.com	Verifies whether domain names are being resolved to IP addresses correctly.
Mail server not reachable	telnet mail.server.com 25	Tests connectivity to the mail server's SMTP port.
Unknown route problem	tracert google.com	Shows the path (hops) packets take and identifies where communication fails.

Bonus Challenge

Q19. Investigate your own machine and answer:

- What is your public IP address?

  IPv4 Address **180.235.122.218**
- What is your private IP address?
  IPv4 Address. . . . . : 11.12.19.76

- Why are they different?

- Private IP Address: Used inside your local network (home, office, college).
- Public IP Address: Assigned by your ISP and visible on the internet.

- They are different because a router uses NAT (Network Address Translation) to allow multiple devices with private IPs to share a single public IP.

Q20. Which command was most useful for troubleshooting and why? Explain with a real example from today's experiment.

The ping command was the most useful for troubleshooting because it quickly checks network connectivity and response time. In today's experiment, I used ping google.com and received replies with 0% packet loss and an average RTT of about 24 ms, confirming that the internet connection was working properly.